An Iterative GLRT for Hyperspectral Target Detection Based on Spectral Similarity and Spatial Connectivity Characteristics

Liang Chen, Jun Liu, *Senior Member, IEEE*, Siyu Sun, Weidong Chen, *Member, IEEE*, Bo Du, *Senior Member, IEEE*, and Rong Liu

***Abstract*— Recently, a generalized likelihood ratio test (GLRT)-based multipixel target detector for hyperspectral imagery (HSI) was proposed. With joint exploitation of the pixels occupied by a target of interest, the detection performance was significantly improved. However, it still faces a pixel selection problem in practice. In this article, we address the pixel selection problem for the multipixel target detector in practice. First, we propose an adaptive target pixel selection method based on spectral similarity and spatial connectivity characteristics. Second, we propose a method to collect the pixels spatially closest to the target pixels as the training background pixels, so that their residual background components share the same statistical characteristics with high probability. To exclude potential target pixels in the collected training background pixels, an iterative version of the GLRT-based multipixel target detector is proposed. It is easy to set the key parameters of the proposed method, which is attractive in practice. Experimental results on four real hyperspectral datasets show that the proposed method outperforms its counterparts in terms of detection performance.**

***Index Terms*— Adaptive pixel selection, hyperspectral imagery (HSI), iterative generalized likelihood ratio test (GLRT), multipixel target, target detection.**

I. Introduction

HYPERSPECTRAL sensors are the combination of spectroscopy and imaging technology. They provide images of the same scene with hundreds of continuous spectral bands. Thus, the hyperspectral imagery (HSI) contains both the spatial and spectral information of the materials in the scene. The abundant information make it possible to distinguish different materials in the scene [1], [2], [3]. Target detection is one of the most important applications of hyperspectral data, which aims to separate target pixels from the background with known target spectral signature [4], [5], [6], [7], [8].

Researchers have developed many solutions to the target detection problem for HSI based on the spectral information, including statistical methods, filter-based methods, representation-based methods, and so on. Statistical methods treat the target and background pixels as random vectors subjecting to specific probability distributions and detect the target pixels by resorting to the hypothesis test theory. Based on different probability distributions, various detectors have been proposed [9], [10], [11]. The adaptive coherence/cosine estimator [12], generalized likelihood ratio test (GLRT) [13], [14], and orthogonal subspace projection (OSP) detector [15], [16] are well-known statistical methods and have been successfully applied in real target detection tasks [12]. Different from statistical methods, filter-based methods make no assumption on the probability distributions of the pixels. Filter-based target detection methods try to find a filter, which minimizes a specific statistic of the detection outputs [17], [18]. The constrained energy minimization (CEM) [17], which fixes the output of the desired target signature while minimizing the second-order statistic of the detection outputs, has been widely adopted for hyperspectral target detection. Representation-based detection methods assume that a target (or background) pixel can be approximated by a linear combination of the elements in the target (or background) dictionary [4], [19], [20], [21]. By comparing the approximate errors, the pixel under test is declared to be a target pixel or a background pixel. The sparse representation and collaborative representation methods are widely studied representation-based methods for target detection in HSI [22], [23], [24].

When the spatial resolution of the hyperspectral sensor is smaller than the size of targets, targets occupy multiple consecutive pixels (called multipixel targets) in HSI [25], [26], [27], [28]. The additional spatial information provides a potential way to improve the detection performance of multipixel targets. Many detection methods have been proposed by jointly exploiting the spatial and spectral information. In [29], a Sparse-SpatialCEM was proposed by introducing the total variation to smooth the detection outputs of CEM. In [30], [31], and [32], the Laplacian and joint sparsity are introduced to incorporate the spatial correlation of neighboring pixels by limiting their representation coefficient vectors. The

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detection performances of these methods have shown much improvement by introducing spatial correlation of neighboring pixels. A practical problem for these methods is that they do not have a constant false alarm rate (CFAR) property, which is highly desirable for a detection scheme [25], [33], [34]. Their detection thresholds are usually determined after all detection outputs are obtained. Tensor representation technology has been introduced by reforming each pixel and its surrounding pixels as a small tensor block to capture the inner correlation among spatial and spectral dimensions [35], [36], [37]. In [35], by assuming that the target and background tensor blocks lie in different low tensor subspaces, the tensor matched subspace detector was proposed. In [36], the tensor sparse representation technology is introduced for hyperspectral target detection. To implement these methods, the target tensor subspaces or training target tensors are usually required as a priori information, which may not be available in practice.

In [26], each pixel in the HSI was assumed to have a multidimensional Gaussian distribution, and a multipixel CFAR detector based on the GLRT criterion was derived with known target spectral signature and spatial pattern. The spatial pattern means the positions and the abundances of the pixels occupied by a target. In reality, the spatial pattern of the multipixel target may be unknown, since the position and abundance of each pixel of the target may vary at different viewing angles and illumination during hyperspectral imaging. To detect multipixel targets with unknown spatial pattern, a GLRT-based multipixel target detector was proposed recently in [38]. However, in practice, the detector performs no good at low probability of false alarm (PFA) by adopting the typical dual window scheme [39], [40] to extract target pixels and training background pixels. The main reason is that there exists too much spatial shape mismatch between the target (background) pixels and the inner window of the typical dual window scheme.

In this article, we address the problem of detecting the multipixel target with an unknown spatial pattern in Gaussian background for HSI. We adopt the idea of the GLRT-based multipixel target detector [38] and propose its iterative version, which focuses on selecting suitable target pixels and training background pixels in practice. Specifically, we propose an adaptive target pixel selection method to select the target pixels, based on the spectral similarity and spatial connectivity characteristics. The pixels spatially closest to the selected target pixels are used as the training background pixels to estimate the covariance matrix. To alleviate the influence of target pixels blended in training background pixels, an iterative version of the GLRT-based multipixel target detector is proposed. Experimental results on four real hyperspectral datasets show that the proposed detector outperforms its counterparts.

The remainder of this article is organized as follows. In Section II, the GLRT-based multipixel detector is introduced, and the motivation is presented. In Section III, an iterative GLRT is proposed for collecting target pixels and training background pixels. Experiment results on four real hyperspectral datasets are illustrated in Section IV. Finally, this article is concluded in Section V.

Notation: Vectors (matrices) are denoted by boldface lower (upper) case letters. Superscript $(\cdot)^T$ denotes transpose. The notation $\sim$ means "is distributed as," and $\mathcal{N}(\boldsymbol{\mu}, \mathbf{R})$ denotes a multidimensional real-valued Gaussian distribution with mean $\boldsymbol{\mu}$ and covariance matrix $\mathbf{R}$. $\mathbf{0}_{m\times n}$ denotes an $m \times n$ matrix whose entries are zeros. $\mathbf{I}_p$ stands for a p-dimensional identity matrix. vec$(\cdot)$ stands for the vectorization operator, and $\otimes$ is the Kronecker product. The symbol $\mathbf{C} \sim \mathcal{N}(\mathbf{M}, \mathbf{R}_n \otimes \mathbf{R}_m)$ denotes that the random matrix $\mathbf{C} \in \mathbb{R}^{n\times m}$ has a matrix variate normal distribution with mean matrix $\mathbf{M} \in \mathbb{R}^{n\times m}$ and covariance matrix $\mathbf{R}_n \otimes \mathbf{R}_m$, where $\mathbf{R}_n \in \mathbb{R}^{n\times n}$ and $\mathbf{R}_m \in \mathbb{R}^{m\times m}$, if vec$(\mathbf{C}^T) \sim \mathcal{N}(\text{vec}(\mathbf{M}^T), \mathbf{R}_n \otimes \mathbf{R}_m)$. $|\cdot|$ is the module of a vector. det$(\cdot)$ and tr$(\cdot)$ denote the determinant and trace of a matrix, respectively. $\Gamma(\cdot)$ is the Gamma function. card$(\cdot)$ denotes the cardinality of a set. $\lfloor\cdot\rfloor$ is the floor function.

II. GLRT-Based Multipixel Target Detector

The multipixel target detection problem can be formulated as a binary hypothesis test to decide between the null hypothesis H_0 and the alternative one H_1

$$H_0 : \begin{cases} \mathbf{y}_j = \mathbf{n}_j, & j = 1, 2, \ldots, J \\ \mathbf{y}_k = \mathbf{n}_k, & k = J+1, J+2, \ldots, J+K \end{cases} \tag{1a}$$

and

$$H_1 : \begin{cases} \mathbf{y}_j = \mathbf{p}a_j + \mathbf{n}_j, & j = 1, 2, \ldots, J \\ \mathbf{y}_k = \mathbf{n}_k, & k = J+1, J+2, \ldots, J+K \end{cases} \tag{1b}$$

where the values of $\mathbf{y}_j \in \mathbb{R}^{N\times 1}, j = 1, 2, \ldots, J$, are the observed pixels containing potential targets (target pixels), the values of $\mathbf{y}_k \in \mathbb{R}^{N\times 1}, k = J+1, J+2, \ldots, J+K$, are the training background pixels, $\mathbf{n}_j, j = 1, 2, \ldots, J$, and $\mathbf{n}_k, k = J+1, J+2, \ldots, J+K$, are the residual background component assumed to be independent identically distributed as multidimensional real-valued Gaussian distribution with zero mean and unknown covariance matrix $\mathbf{R}$, $\mathbf{p}a_j$ stands for the target component in $\mathbf{y}_j$, $\mathbf{p}$ denotes the known target signature, and a_j is a deterministic but unknown scalar corresponding to the target signature abundance in the jth target pixel.

We denote the target pixels and training background pixels by

$$\mathbf{Y}_J = \left[\mathbf{y}_1, \mathbf{y}_2, \ldots, \mathbf{y}_J\right] \tag{2}$$

and

$$\mathbf{Y}_K = \left[\mathbf{y}_{J+1}, \mathbf{y}_{J+2}, \ldots, \mathbf{y}_{J+K}\right] \tag{3}$$

respectively. The binary hypothesis test (1) can be recast as follows:

$$H_0 : \begin{cases} \mathbf{Y}_J \sim \mathcal{N}(\mathbf{0}_{N\times J}, \mathbf{R} \otimes \mathbf{I}_J) \\ \mathbf{Y}_K \sim \mathcal{N}(\mathbf{0}_{N\times K}, \mathbf{R} \otimes \mathbf{I}_K) \end{cases} \tag{4a}$$

and

$$H_1 : \begin{cases} \mathbf{Y}_J \sim \mathcal{N}\left(\mathbf{p}\mathbf{a}^T, \mathbf{R} \otimes \mathbf{I}_J\right) \\ \mathbf{Y}_K \sim \mathcal{N}(\mathbf{0}_{N\times K}, \mathbf{R} \otimes \mathbf{I}_K) \end{cases} \tag{4b}$$

where $\mathbf{a} = [a_1, a_2, \ldots, a_J]^T$ is the unknown abundance vector of the target signature.

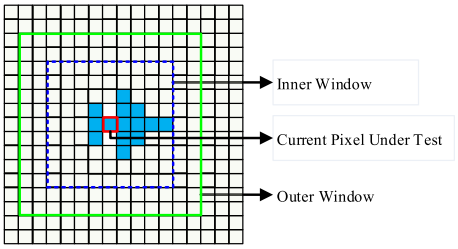


Fig. 1. Spatial dual window scheme: outer window (the green solid square) and inner window (the blue dashed square).

For the binary hypothesis test problem in (4), a GLRT-based multipixel detector was derived in [38] as follows:

$$\eta = \frac{\mathbf{p}^T\mathbf{S}^{-1}\mathbf{Y}_J\left(\mathbf{I}_J + \mathbf{Y}_J^T\mathbf{S}^{-1}\mathbf{Y}_J\right)^{-1}\mathbf{Y}_J^T\mathbf{S}^{-1}\mathbf{p}}{\mathbf{p}^T\mathbf{S}^{-1}\mathbf{p}} \underset{H_0}{\overset{H_1}{\gtrless}} \lambda \tag{5}$$

where

$$\mathbf{S} = \mathbf{Y}_K\mathbf{Y}_K^T. \tag{6}$$

λ is a predefined detection threshold. As given in [38], the PFA for the GLRT-based multipixel detector can be written as follows:

$$P_{\text{FA}} = 1 - \int_0^{\lambda'} \frac{1}{B\left(\frac{J}{2}, \frac{K-N+1}{2}\right)} \frac{\eta'^{J/2-1}}{(1+\eta')^{(J+K-N+1)/2}} d\,\eta' \tag{7}$$

where

$$B(a,b) = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)} \tag{8}$$

and

$$\lambda' = \frac{\lambda}{1-\lambda}. \tag{9}$$

It can be seen that the expression of the PFA is independent of the covariance matrix, which indicates its CFAR property against the background covariance matrix. The CFAR property is highly desirable for a detection scheme, since it implies that the test statistic does not depend on the variation of the background statistics, so that a unique test threshold can be set to obtain the same false alarm rate all over the image.

In practice, the target pixels and training background pixels are usually extracted from the neighborhood of the current pixel under test. In [38], the widely used dual window scheme was adopted. As shown in Fig. 1, the neighborhood of the current pixel under test is parted into two regions: a small inner window region centered within a large outer window region. The inner window is used to enclose potential target pixels, and the pixels in the inner window are selected as target pixels ($\mathbf{Y}_J$). The pixels between the inner and outer windows are collected as training background pixels ($\mathbf{Y}_K$). When the value of the detector exceeds a predefined threshold, the central pixel is declared as a target pixel. However, the dual window scheme may cause some problems. One is the mismatch problem between the spatial shapes of the inner window and the target (or background) pixels. Due to the mismatch, the selected $\mathbf{Y}_J$ may be contaminated by too many background (or target) pixels, which deteriorates the detection performance. The other one is that target pixels may be blended in $\mathbf{Y}_K$, which reduces the precision of the estimated background covariance matrix and deteriorates the detection performance. To deal with these problems, we proposed an adaptive strategy to extract the target pixels $\mathbf{Y}_J$ and training background pixels $\mathbf{Y}_K$ in Section III.

III. Proposed Method

A. Target Pixels Selection

In HSI, the pixels belonging to the same class usually have similar spectral signatures (referred to spectral similarity characteristic). Since the multipixel target occupies several consecutive pixels in HSI, when we treat the HSI as a graph where the pixels are the vertices and there exists edges between each pixel and its spatially nearest eight neighbors, the target pixels should be connected (called connectivity characteristic). We can adaptively collect the target pixels by resorting to the two characteristics. We adopt the angle distance (AD) [41] to measure the similarity between two spectral vectors, which is defined as follows:

$$\text{AD}(\mathbf{z}_1, \mathbf{z}_2) = \arccos\left(\frac{\mathbf{z}_1^T\mathbf{z}_2}{\sqrt{\mathbf{z}_1^T\mathbf{z}_1}\sqrt{\mathbf{z}_2^T\mathbf{z}_2}}\right). \tag{10}$$

The smaller the angle, the higher the similarity between them. To extract the target pixels $\mathbf{Y}_J$ for a given J, we formulate the target pixel selection problem based on these two characteristics as follows:

$$\begin{aligned}\mathbf{Y}_J(\mathbf{y}_0) = &\arg\min_{\mathbf{Y}_J} \sum_{\mathbf{y}_i \in \mathbf{Y}_J} \text{AD}(\mathbf{y}_0, \mathbf{y}_i) \\ &\text{s.t. } \mathbf{y}_0 \in \mathbf{Y}_J, \text{ and } \mathbf{Y}_J \text{ are connected.}\end{aligned} \tag{11}$$

All connected pixel sets that include $\mathbf{y}_0$ should be calculated to obtain the optimal solution to the above problem, which means heavy computational burden. Here, we proposed a suboptimal solution with low computational complexity by selecting the target pixels one by one in an iterative way. The initially selected target pixels $\mathbf{Y}_J^0$ are set to only contain the current pixel under test ($\mathbf{y}_0$). In the ith iteration, the set of candidates $\mathbf{Y}_C^i$ consists of all the pixels that are connected to the selected target pixels $\mathbf{Y}_J^{i-1}$ [i.e., $\mathbf{Y}_C^i = \Omega_C(\mathbf{Y}_J^{i-1})$]; then, one pixel in $\mathbf{Y}_C^i$ that is the most similar to $\mathbf{y}_0$ is chosen as a newly selected target pixel in $\mathbf{Y}_J^i$. The iteration ends when J target pixels are selected. Then, we get a set of connect target pixels that are vary similar to the current pixel under test.

B. Training Background Pixels Selection

The training background pixels should be spatially located as close to the target pixels as possible, so that their residual background component share the same distribution with high probability. We propose an iterative strategy to extract the training background pixels for a given number K. The input HSI is denoted as $\mathbf{X} \in \mathbb{R}^{H\times W\times N}$, where H and W are the spatial sizes of $\mathbf{X}$. Before the selection proceeds, the pixels connected to the selected target pixels $\mathbf{Y}_J$ are selected as guard pixels to exclude potential target pixels in the training background pixels. The initial set of selected pixels $\mathbf{Y}_S^0$ is selected as the union set of $\mathbf{Y}_J$ and the guard pixels. In the

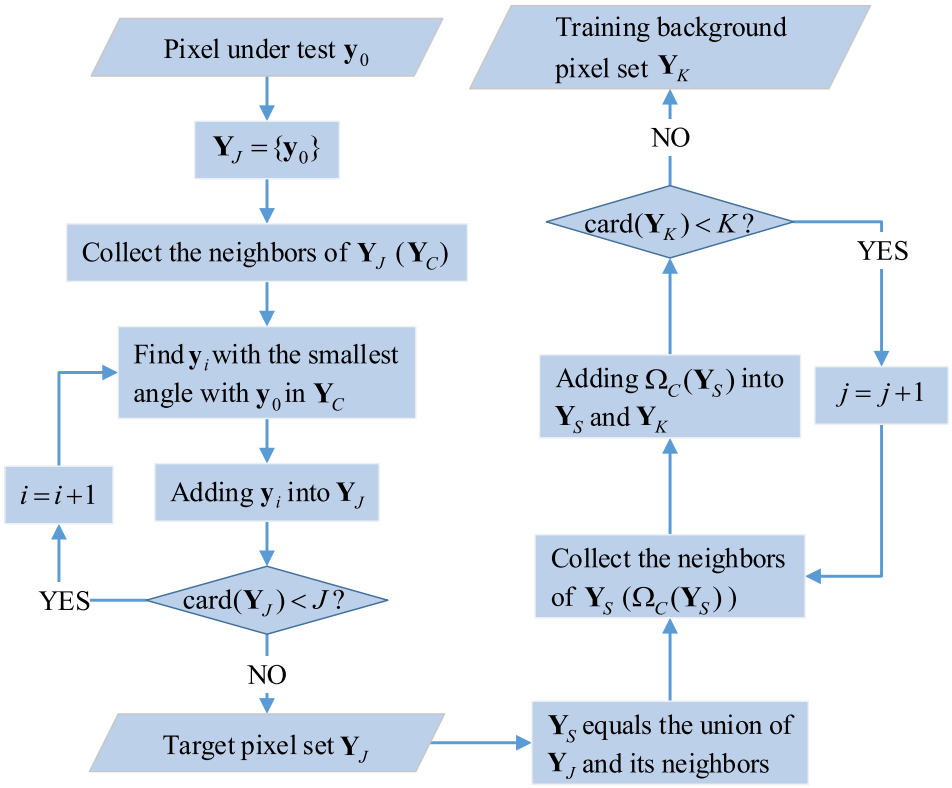


Fig. 2. Flowchart of the adaptive pixels selection.

jth iteration, the pixels connected to the selected pixels $\mathbf{Y}_S^{j-1}$ are collected as new training background pixels and labeled as selected pixels [i.e., $\mathbf{Y}_K^j = \mathbf{Y}_K^{j-1} \cup \Omega_C(\mathbf{Y}_S^{j-1})$ and $\mathbf{Y}_S^j = \mathbf{Y}_S^{j-1} \cup \Omega_C(\mathbf{Y}_S^{j-1})$]. The iteration ends when K training background pixels are obtained.

Given the numbers of the target pixels and the training background pixels (i.e., J and K), the step-by-step implementation of the adaptive pixel selection is provided in Algorithm 1. A flowchart of the adaptive pixel selection procedure is demonstrated in Fig. 2. In practice, the two parameters J and K can be set as follows: with known target size $L_T \times W_T$ and spatial resolution of the sensor $L_R \times W_R$, and J can be set to be

$$J = \left\lfloor \frac{L_T \times W_T}{L_R \times W_R} \right\rfloor. \tag{12}$$

Also, K is set according to the number of the spectral bands N. Usually, $K = 2N$ can ensure satisfactory accuracy in the background covariance matrix estimation.

Algorithm 1 Adaptive Pixels Selection Procedure

Input: The hyperspectral image data $\mathbf{X}$, the current pixel under test $\mathbf{y}_0$, the number of the target pixels J, and the number of the training background pixels K.
Output: $\mathbf{Y}_J$ and $\mathbf{Y}_K$.
Initialization: Set $\mathbf{Y}_J^0 = \mathbf{y}_0$, $\mathbf{Y}_K^0 = \emptyset$, $N_{ks}^0 = 0$, and $j = 0$.
for $i = 1 : J-1$ **do**
  $\mathbf{Y}_C^i = \Omega_C(\mathbf{Y}_J^{i-1})$;
  $\mathbf{y}_i = \min_{\mathbf{y} \in \mathbf{Y}_C^i} \mathrm{AD}(\mathbf{y}, \mathbf{y}_0)$;
  $\mathbf{Y}_J^i = \mathbf{Y}_J^{i-1} \cup \mathbf{y}_i$;
Set $\mathbf{Y}_S^0 = \mathbf{Y}_J^{J-1} \cup \Omega_C(\mathbf{Y}_J^{J-1})$;
while $N_{ks}^j < K$ **do**
  $j = j + 1$;
  $\mathbf{Y}_K^j = \mathbf{Y}_K^{j-1} \cup \Omega_C(\mathbf{Y}_S^{j-1})$;
  $\mathbf{Y}_S^j = \mathbf{Y}_S^{j-1} \cup \mathbf{Y}_K^j$;
  $N_{ks}^j = \mathrm{card}(\mathbf{Y}_K^j)$;
$\mathbf{Y}_J = \mathbf{Y}_J^{J-1}$ and $\mathbf{Y}_K = (\mathbf{Y}_K^j)_K$, where $(\mathbf{Y}_K^j)_K$ means the first K selected pixels in $\mathbf{Y}_K^j$.

C. Iterative GLRT

In Section III-B, the training background pixels are selected spatially as close to the selected target pixels as possible. However, there may exist some target pixels blended in the training background pixels. A possible case is demonstrated in Fig. 3. As discussed in [42] and [43], the contaminated training background pixels obviously deteriorate the detection performance. To tackle this problem, similar to [44], the detection results can be utilized to locate and remove the potential target pixels in the set of training background pixels. Based on this idea, we propose an iterative version of multipixel GLRT method to purify the set of training background pixels. The initial set of potential target pixels is $\mathbf{Y}_P^0 = \emptyset$. In the ith iteration, after excluding the pixels in $\mathbf{Y}_P^{i-1}$ from the training

Algorithm 2 Procedure of the Proposed Method

Input: The hyperspectral image data $\mathbf{X}$, the target signature $\mathbf{p}$, P_{FA}, the number of the target pixels J, the number of the training background pixels K, and the maximum iteration number N_I.
Output: The detection image DI.
Initialization: Set $\mathbf{Y}_P^0 = \emptyset$.
Calculate λ by (7);
for $i = 1 : N_I$ **do**
  for each pixel $\mathbf{X}(h, w, :)$ **do**
    Select $\mathbf{Y}_J$;
    Select $\mathbf{Y}_K$ after excluding pixels in $\mathbf{Y}_P^{i-1}$;
    Calculate $\boldsymbol{\eta}^i(h, w)$ applying (5);
  $\mathbf{Y}_P^i = \{\mathbf{X}(h, w, :) \mid \boldsymbol{\eta}^i(h, w) > \lambda\}$;
  if $\mathbf{Y}_P^i = \mathbf{Y}_P^{i-1}$ **then**
    break;
DI $= \boldsymbol{\eta}^i$.

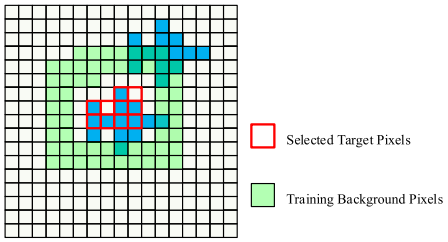


Fig. 3. Case of target pixels blended in the training background pixels.

background pixels, the test statistics are calculated according to (5) to get the detection outputs $\boldsymbol{\eta}^i$. Then, the set of potential target pixels is updated to be $\mathbf{Y}_P^i = \{\mathbf{X}(h, w, :) \mid \boldsymbol{\eta}^i(h, w) > \lambda\}$. When $\mathbf{Y}_P^i = \mathbf{Y}_P^{i-1}$, the iteration ends, and the detection image (DI) is denoted as $\boldsymbol{\eta}^i$.

The step-by-step implementation of the proposed method is presented in Algorithm 2.

In Fig. 4, a flowchart describes the implementation of the proposed iterative multipixel GRLT method.

IV. Hyperspectral Data Experiments

To verify the effectiveness of the proposed method, we compare the proposed method with three detectors, including the GLRT [13], the target detection method based on constrained sparse representation and binary hypothesis model (CSRBBH) [24], and the Sparse-SpatialCEM [29]. As in [5], [45], and [46], the receiving operating characteristic (ROC) curve is used for performance assessment. The ROC curve plots the detection probability as a function of PFA to provide a threshold-free performance comparison. A closer curve to the top-left corner means better detection performance. The area under the curve (AUC) value calculates the area enclosed by the curve and the PFA axis to provide a quantitative measurement of ROC.

A. Hyperspectral Datasets

The proposed method is evaluated on four hyperspectral datasets. The details are listed in the following.

The first hyperspectral dataset was collected by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor over the Los Angeles area [47]. There are 205 spectral bands after removing bad bands (i.e., the water absorption and low signal-to-noise ratio bands). The two aircrafts with 87 pixels are the intended targets to be detected in the image. The dataset and its ground truth are demonstrated in Fig. 5, and the spatial size is 100×100.

The second hyperspectral dataset was acquired by the AVIRIS sensor [48] over Indian Pines test site in Northwest Indian. There are 202 spectral bands after removing bad bands. There are 16 classes of land covers, and the alfalfa (class 1) containing 54 pixels is selected as the target to be detected in the image. The dataset and its ground truth are depicted in Fig. 5, and the spatial size is 145×145.

The third hyperspectral dataset was obtained by the AVIRIS sensor over the Gulfport area [47]. There are 191 spectral bands after removing bad bands. The three aircraft covering 60 pixels are chosen to be the targets to be detected in the image. The dataset and its ground truth are delineated in Fig. 5, and the spatial size is 100×100.

The fourth hyperspectral dataset was captured by the AVIRIS sensor over the San Diego airport area, CA, USA [5]. There are 189 spectral bands after removing bad bands. The three aircrafts containing 134 pixels are regarded as the target to be detected in the image. The dataset and its ground truth are illustrated in Fig. 5, and the spatial size is 400×400.

B. Experimental Settings

In our experiments, some parameters are described as follows. As in [45] and [49], we select several pixels from the targets as the target signatures in each of the datasets. The a priori target signature is set to be the mean of the selected target signatures for the GLRT, Sparse-SpatialCEM, and the proposed method. The training background pixels are acquired by the way introduced in Section III-B. The GLRT and CSRBBH share the same training background pixels in each dataset. The number of the training background pixels K for each dataset is set to be twice the amount of the spectral bands. For the CSRBBH, the selected target signatures and the training background pixels consist of the atoms of the target and background dictionaries, respectively. The contamination level is tuned among 0, 0.02, and 0.05 to report the best detection performance, which means the AUC value reaches maximum. For the Sparse-SpatialCEM, we set $\mu = 0.7$, $\lambda = 1$, $\rho_1 = 0.4$, and $\rho_2 = 0.4$, as in [37]. For the proposed method, the numbers of the target pixels J are set to be 40, 50, 11, and 40, respectively, for the four datasets according to the size of a single target in each dataset. During the iterative procedure of the proposed method, P_{FA} is set to be 10^{-5} for all of the datasets.

C. Experimental Results and Analysis

The 2-D plots of the detection results are shown in Fig. 5. For the Los Angeles dataset, it is shown that more target pixels can be distinguished from the background by the proposed method, which indicates that the proposed method

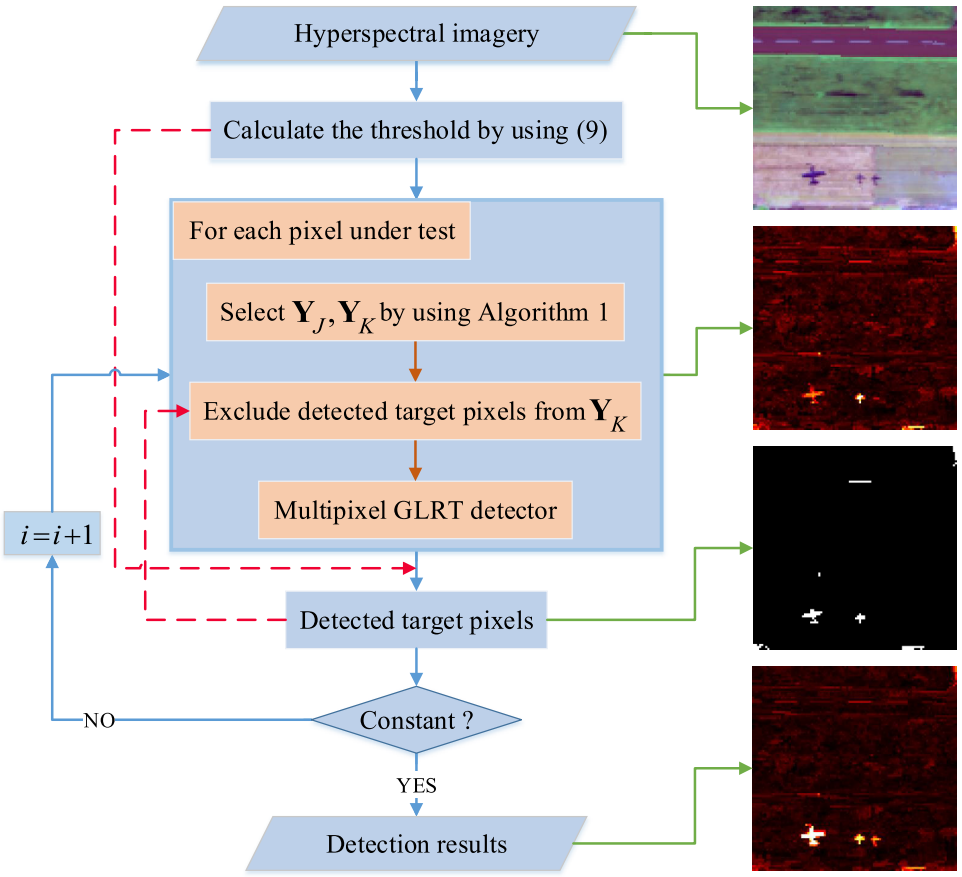


Fig. 4. Flowchart of the proposed method.

TABLE I
AUC VALUE WITH 95% CONFIDENCE INTERVALS ON PFA COMPARISONS ON THE FOUR DATASETS

Datasets	GLRT [13]	CSRBBH [24]	Sparse-SpatialCEM [29]	Proposed
Los Angeles	$0.79908^{+0.00391}_{-0.00411}$	$0.74937^{+0.00284}_{-0.00298}$	$0.97517^{+0.00199}_{-0.00228}$	$\mathbf{0.99328^{+0.00072}_{-0.00105}}$
Indian Pines	$0.92401^{+0.00143}_{-0.00156}$	$0.95261^{+0.00229}_{-0.00241}$	$0.97606^{+0.00078}_{-0.00092}$	$\mathbf{0.99801^{+0.00038}_{-0.00054}}$
Gulfport	$0.86006^{+0.00386}_{-0.00408}$	$0.56575^{+0.00429}_{-0.00430}$	$0.99610^{+0.00063}_{-0.00097}$	$\mathbf{0.99944^{+0.00021}_{-0.00056}}$
San Diego	$0.85633^{+0.00087}_{-0.00089}$	$0.60513^{+0.00021}_{-0.00021}$	$\mathbf{0.97227^{+0.00068}_{-0.00070}}$	$0.97127^{+0.00024}_{-0.00026}$

has higher probability of detection with the same false alarms. For the Indian Pines, Gulfport, and San Diego datasets, the Sparse-SpatialCEM and the proposed method provide more distinguishable detection results, which verifies the improvement of the target detection performance by including spatial information. However, the Sparse-SpatialCEM gets higher detection values for the background pixels compared with the proposed method, which means that the proposed method suppresses the background more effectively. We can conclude that the proposed method shows the best detection performance.

The ROC curves with 95% confidence intervals on PFA [50] are plotted in Fig. 6. For the Los Angles and Gulfport datasets, it can be observed that the ROC curve of the proposed method is always above those of the other methods. For the Indian Pines dataset, the ROC curve of the proposed method is over those of the other methods on most PFA values. For the San Diego dataset, the ROC curve of the proposed method is over those of the other methods in the low-PFA region, which is the region of greater interest in practice. For all of the four datasets, the proposed detector achieves the best performance in terms of target detection. It can also be seen that the proposed method outperforms the typical pixelwise GLRT. That is attributed to the joint exploitation of multiple pixels occupied by the target.

The AUC values with 95% confidence intervals on PFA of different detectors are calculated and listed in Table I. The proposed method has the largest AUC values for three of the datasets and the second largest AUC value for the San Diego dataset. It shows the superiority of the proposed method in the target detection task.

We vary the parameters J and K of the proposed method to examine their influences on the detection performance. The parameter J is set to range from 50% to 150% of the size of

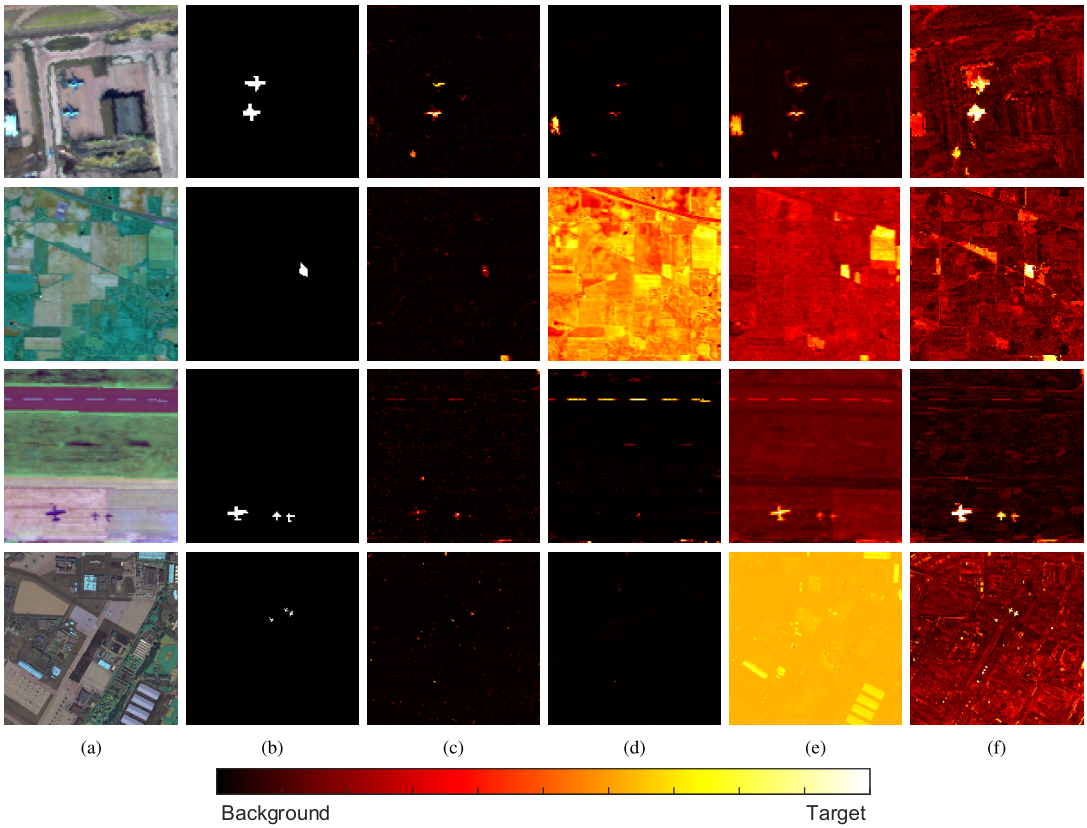


Fig. 5. Two-dimensional plots of the detection results. (Top to bottom) Los Angeles dataset, Indian Pines dataset, Gulfport dataset, and San Diego dataset. (a) False color composite of the image. (b) Ground truth. (c) GLRT. (d) CSRBBH. (e) Sparse-SpatialCEM. (f) Proposed.

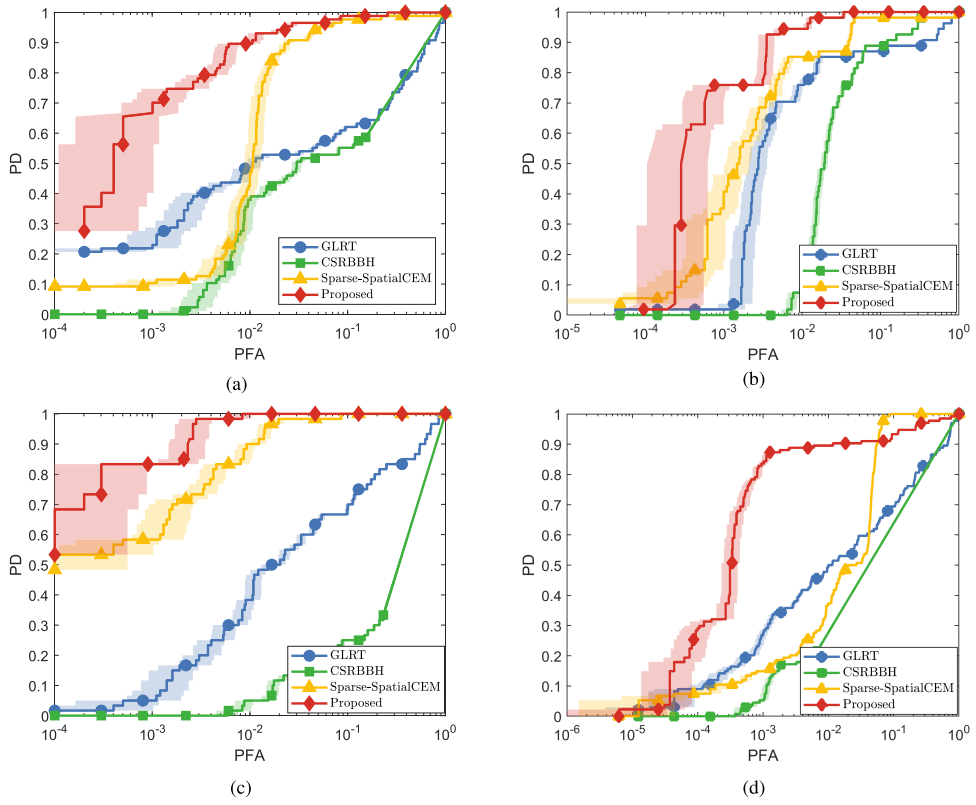


Fig. 6. ROC curves of the compared methods. (a) Los Angeles. (b) Indian Pines. (c) Gulfport. (d) San Diego.

the target, and the parameter K is set to range from $1.1N$ to $9N$, where N is the number of the spectral bands of the HSI. The results are illustrated in Fig. 7. It is shown that the AUC value tends to be stable in a wide range of parameters J and K, which indicates that the proposed method is robust to the settings of J and K.

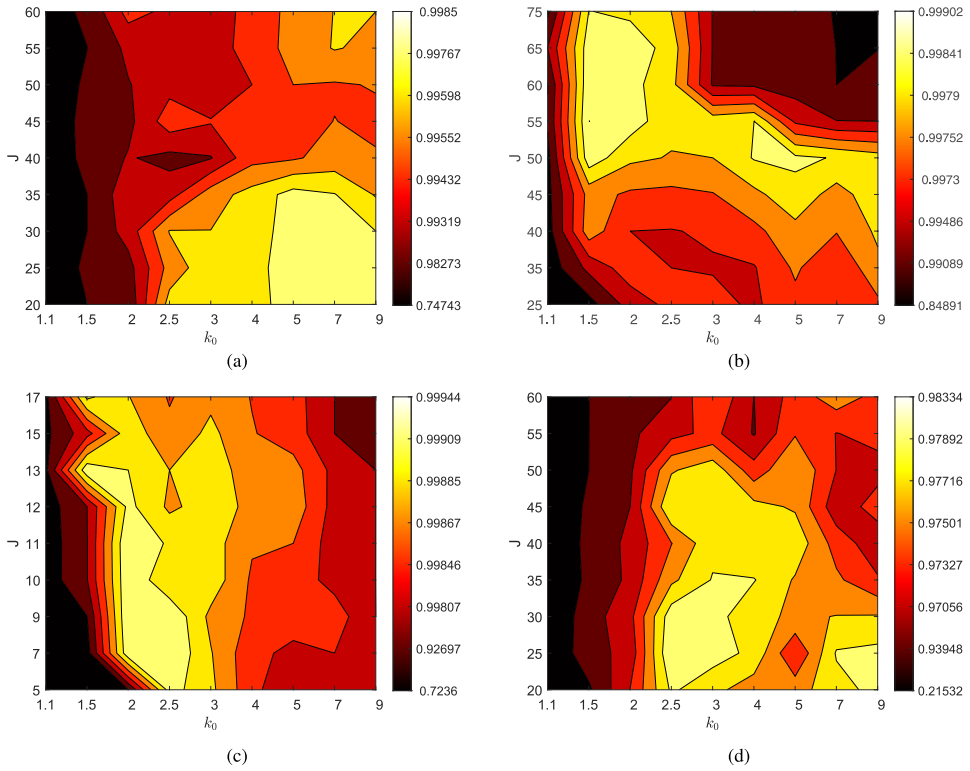


Fig. 7. AUC values under different settings of J and K with $K = k_0 N$. (a) Los Angeles. (b) Indian Pines. (c) Gulfport. (d) San Diego.

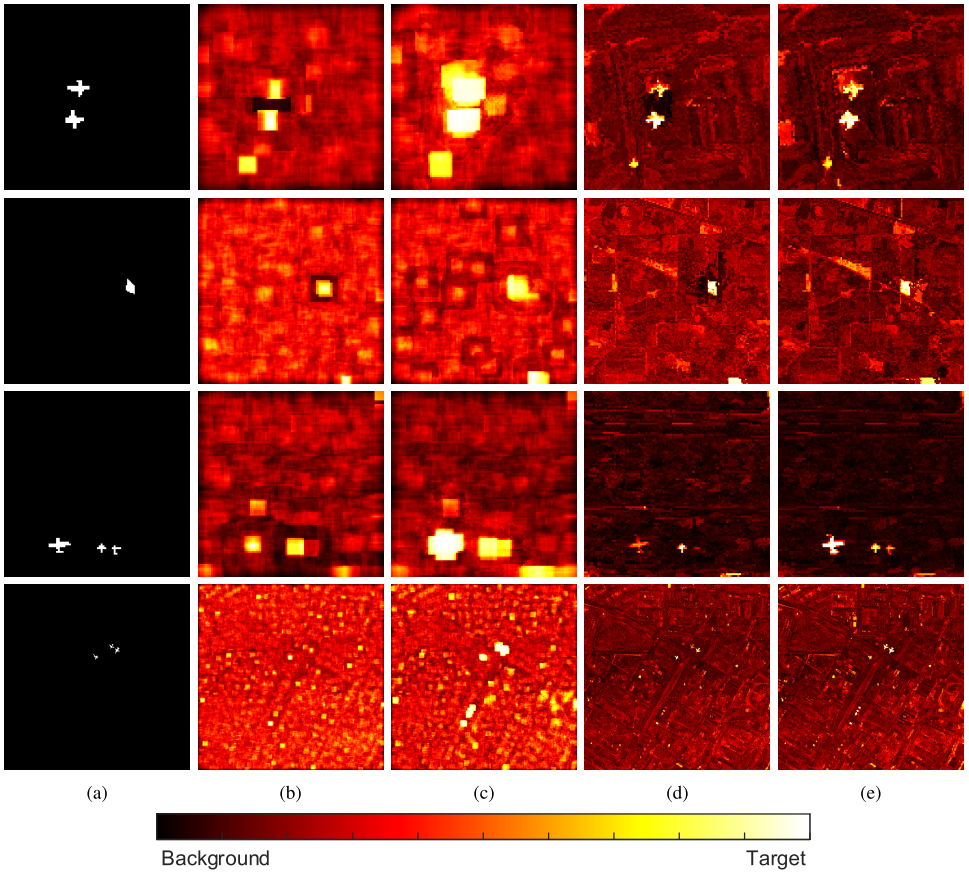


Fig. 8. Two-dimensional plots of the detection results. (Top to bottom) Los Angeles dataset, Indian Pines dataset, Gulfport dataset, and San Diego dataset. (a) Ground truth. (b) DW1-multipixel GLRT. (c) DW2-multipixel GLRT. (d) PS-multipixel GLRT. (e) Proposed.

The proposed method consists of two steps to improve the target detection performance. One is the adaptive selection of proper target/training background pixels, and the other is the elimination of background contamination. To investigate the contribution of each step to the performance improvement, four detectors are as follows: 1) the multipixel GLRT

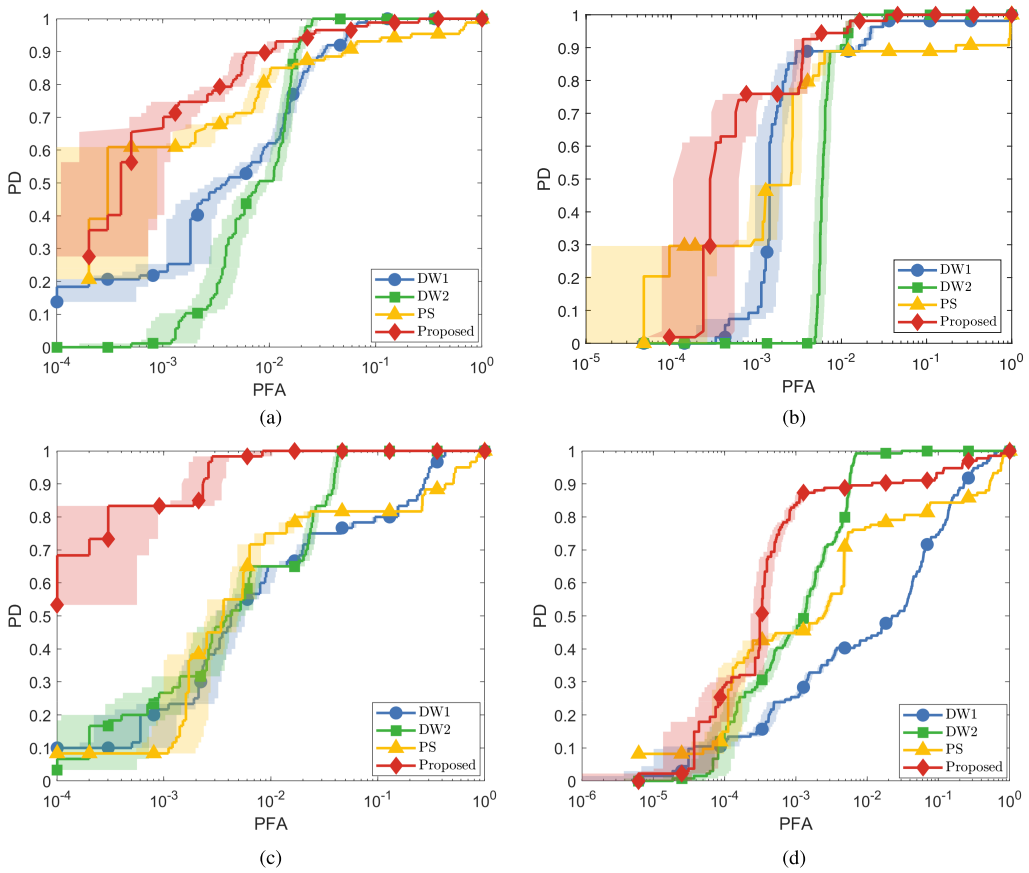
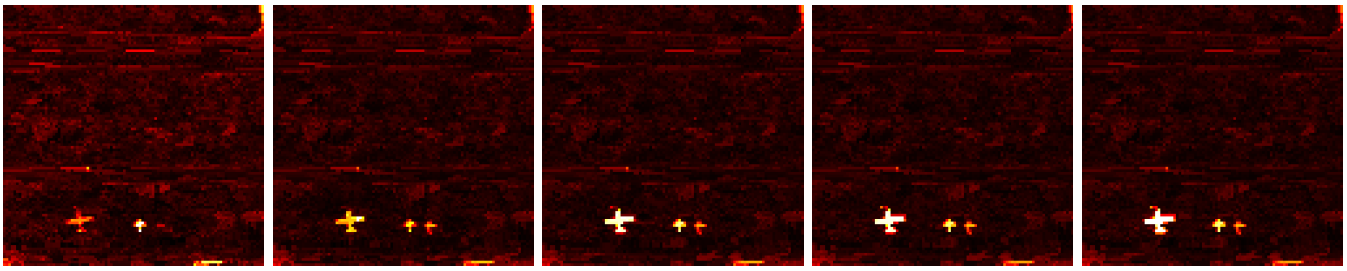


Fig. 9. ROC curve comparisons. (a) Los Angeles. (b) Indian Pines. (c) Gulfport. (d) San Diego.

adopting the dual window scheme, called DW1-multipixel GLRT [38]; 2) the multipixel GLRT with the iterative removal of detected target pixels from training background pixels, called DW2-multipixel GLRT; 3) the multipixel GLRT adopting the target/training background pixels selection scheme in Algorithm 1 of Section III, called PS-multipixel GLRT; and 4) the proposed method is conducted on the four datasets. For the DW1-multipixel GLRT and DW2-multipixel GLRT, the sizes of the inner window are set to be 11×11, 11×11, 9×9, and 11×11 for the four datasets, respectively. The sizes of the outer window are set to be 25×25, 23×23, 23×23, and 23×23 for the four datasets to ensure that the number of training background pixels is bigger than twice the number of the spectral bands.

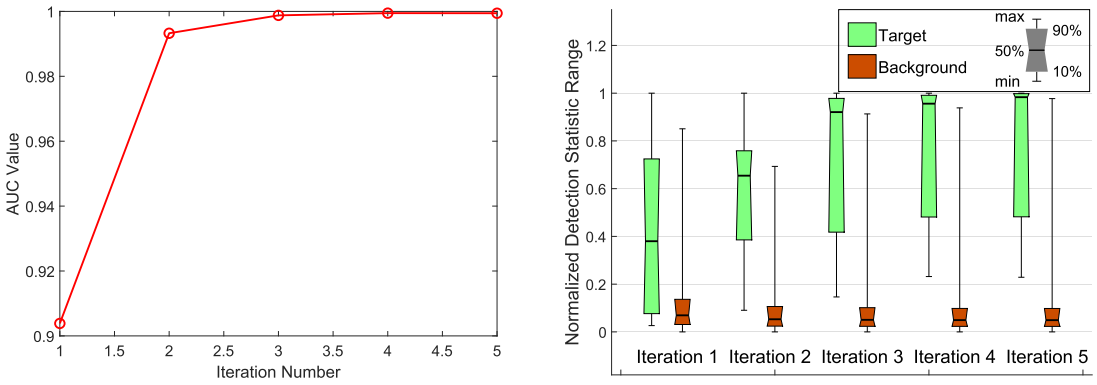
The 2-D plots of the detection results are demonstrated in Fig. 8. It can be observed that the PS-multipixel GLRT and the proposed method show clearer shapes of the targets than the DW1-multipixel GLRT and DW2-multipixel GLRT, which means that the former two detection methods obviously reduce the number of false-alarm pixels around the targets. It is also shown that the PS-multipixel GLRT and the proposed method get lower detection values for the background pixels, which means that they have better performance for background suppression. That is due to the adaptive target/training background pixels selection. In addition, it can be seen that compared with the PS-multipixel GLRT, the proposed method has higher detection values for the target pixels, due to the elimination of background contamination. Compared with the DW1-multipixel GLRT, the DW2-multipixel GLRT also has higher detection values for the target pixels. However, the DW2-multipixel GLRT gets higher detection values for many background pixels (e.g., pixels around the targets) at the same time, which may result in false alarms.

The ROC curves are illustrated in Fig. 9. For the Los Angeles and San Diego datasets, the ROC curve of the PS-multipixel GLRT is above that of the DW1-multipixel GLRT on most PFA values. For the Gulfport dataset, the ROC curves of the PS-multipixel GLRT and DW1-multipixel GLRT are not separated so far as do for the Los Angeles and San Diego datasets. The ROC curve of the PS-multipixel GLRT is still over that of the DW1-multipixel GLRT on most FPA values. However, the ROC curve of the PS-multipixel GLRT is below that of the DW1-multipixel GLRT in the high-PFA region for each of the four datasets. The reason is that the adaptive target pixels selection procedure may not include all pixels of one target, and some of them are treated as training background pixels, which results in background contamination problem. By applying the proposed iterative procedure to remove detected target pixels from training background pixels, the proposed method shows significant improvements for the target detection performance. Based on the above observations, we can conclude that each of the two steps can bring in performance improvements.



The first iteration. The second iteration. The third iteration. The fourth iteration. The fifth iteration.

Fig. 10. Two-dimensional plots of the detection results of the proposed method in different iterations on the Gulfport dataset.



(a) (b)

Fig. 11. Performance of the proposed method in different iterations on the Gulfport dataset. (a) AUC value. (b) Separability map.

The experimental results of the proposed method in different iterations on the Gulfport dataset are plotted in Figs. 10 and 11. It is shown that the detection performance improves, as the number of iterations increases at the beginning, and then tends to be stable. The gap between the target and background boxes obviously grows, as the iteration proceeds. The experimental results on other datasets also exhibit similar phenomena, which are not presented here for brevity. Based on the above results, we can conclude that the detection performance is significantly improved by the proposed iterative version of multipixel GLRT.

V. Conclusion

We have considered the problem of detecting multipixel targets with an unknown spatial pattern for HSI. Multiple pixels occupied by the target were jointly utilized to improve the detection performance. We have focused on the problem of selecting suitable target pixels and training background pixels for the GLRT-based multipixel target detector in practice. To extract the target pixels, an adaptive method based on spectral and spatial characteristics has been proposed. We have also proposed a method to collect the training background pixels spatially as close to the target pixels as possible. Moreover, to alleviate the influence of target pixels blended in training background pixels, the iterative version of the GLRT-based multipixel target detector has been proposed. Experimental results on four real hyperspectral datasets show that the proposed method outperforms its counterparts in terms of detection performance. Specifically, the multipixel GLRT with the proposed target/training background pixels selection scheme gets clearer shapes of the targets. However, it suffers from background contamination problem. By iteratively excluding detected target pixels from training background pixels, the proposed method significantly improves the target detection performance. In addition, the proposed method is robust to the settings of the numbers of the target pixels and the training background pixels (i.e., J and K), which makes it easy to implement in practice. In the future work, we will consider another estimator of the background covariance matrix (e.g., robust estimators) and extend this work to the elliptical distribution case.

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